

Improvements on SadTalker-based Approach for ViCo Conversational Head Generation Challenge

Wei Dai
Robo Space
Beijing, China
loveispdvd@gmail.com

ABSTRACT

This paper presents our solution in the ACM Multimedia ViCo 2023 Conversational Head Generation Challenge, which aims to generate vivid face-to-face conversation videos based on audio and reference images. Our approach builds upon the SadTalker framework with several improvements. Since SadTalker is already a mature and well-developed framework, consisting of multiple stages of algorithms and models, our enhancements mainly focus on the preprocessing and postprocessing stages. With our improved method, we achieved second place in both the Talking Head Generation track and the Listening Head Generation track.

CCS CONCEPTS

• **Computing methodologies** → **Reconstruction.**

KEYWORDS

Digital Human; Conversational Head Generation

ACM Reference Format:

Wei Dai. 2023. Improvements on SadTalker-based Approach for ViCo Conversational Head Generation Challenge. In *Proceedings of the 31st ACM International Conference on Multimedia (MM '23)*, October 29–November 3, 2023, Ottawa, ON, Canada. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3581783.3612866>

1 INTRODUCTION

1.1 Background of the Challenge

In face-to-face communication, participants exchange verbal and non-verbal messages. The speaker conveys their message explicitly, adjusting their pace based on the listener's reactions. The listener provides real-time feedback through non-verbal behaviors. For digital humans engaging in face-to-face human-computer interaction, generating realistic talking head videos and exhibiting appropriate responsive listening behavior is crucial.

This challenge focuses on the extended ViCo dataset[5], the first video conversation dataset encompassing face-to-face dialogue clips across various scenarios. Unlike the previous challenge in 2022, this iteration has expanded its collection of videos to facilitate

the application of more advanced machine learning techniques. The objective is to foster a visual competition centered on generating face-to-face interactive head videos.

Based on this dataset, they have proposed a new benchmark[4] for generating conversation heads in face-to-face interactions. In order to address the limitations of one-way interaction, they have introduced three novel tasks that are more interactive and comprehensive. Through experiments, they have verified that their baseline is capable of generating responsive and realistic agents that can effectively engage in conversations with real individuals.

The ViCo 2023 Conversational Head Generation Challenge consists of the following two tracks:

- **Vivid Talking Head Video Generation:** Participants are tasked with generating talking head videos that accurately capture the identity and audio signals of the speaker.
- **Responsive Listening Head Video Generation:** Participants are required to generate listening head videos that respond to the speaker's behaviors and audio signals in real-time, while considering the identity of the listener.

1.2 Purpose of the Paper

The purpose of this paper is to present the methods we used to achieve second place in both tracks in the ViCo 2023 Challenge. The focus is on several improvements made to the SadTalker framework¹ [3]. Additionally, experimental results are provided, along with a discussion on future work.

The paper is divided into the following sections: Section 2 introduces the SadTalker framework and highlights several improvements made during the competition. Section 3 presents the experimental results and provides an analysis of these results. Section 4 discusses future work and potential directions for further research.

2 METHOD

2.1 Framework

As mentioned in the Title, Abstract, and Section 1.2, the approach utilized in this competition is primarily based on the SadTalker framework with some improvements. SadTalker is a robust framework that generates a corresponding video from a single image and an audio clip. It provides code and models that encompass multiple stages of video generation, making it well-suited for the task requirements of this competition. Therefore, in this section, we will briefly introduce the methods and principles of SadTalker.

SadTalker is a framework that generates realistic talking head videos by utilizing 3D motion coefficients derived from audio input. These coefficients, including head pose and expression, are used to

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
MM '23, October 29–November 3, 2023, Ottawa, ON, Canada

© 2023 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM ISBN 979-8-4007-0108-5/23/10...\$15.00
<https://doi.org/10.1145/3581783.3612866>

¹<https://github.com/OpenTalker/SadTalker>

modulate a novel 3D-aware face render. To ensure accuracy, explicit connections between audio and different types of motion coefficients are established. ExpNet is introduced to accurately learn facial expressions, while PoseVAE synthesizes head motion in various styles. The generated 3D motion coefficients are then mapped to the unsupervised 3D keypoints space of the proposed face render, resulting in the synthesis of high-quality videos. This approach addresses challenges such as unnatural head movement, distorted expression, and identity modification that arise from relying solely on coupled 2D motion fields or 3D information.

The main contributions of SadTalker can be summarized as follows:

- Introduction a novel and comprehensive framework for stylized audio-driven single image talking face animation, which utilizes realistic 3D motion coefficients.
- Presentation of ExpNet and PoseVAE as individual components for learning realistic 3D motion coefficients from audio.
- Proposal of a novel semantic-disentangled and 3D-aware face render, which enables the production of realistic talking head videos.

2.2 Evaluation

The evaluation metrics for this competition consist of three major categories: visual quality, naturalness, and task-specific metrics. The first two categories are applicable to both tracks, while the last one is specific to each track. Each category includes the following metrics²:

- Visual Quality
 - CPBD: image sharpness
 - FID: distance between synthetic and real data distributions
 - SSIM: perceptual similarity of image contrast, luminance and structure
 - PSNR: the error of synthetic data compared to real
 - CSIM: identity preservation from ArcFace
- Naturalness
 - ExpL1: L_1 distance of expression coefficients from 3DMM reconstruction
 - PoseL1: L_1 distance of translation/rotation coefficients from 3DMM reconstruction
- Speaker-specified
 - AVOffset: the synchronization offset of lip motion with input audio
 - AVConf: the confidence of AVOffset
 - LipLMD: the lip landmark distance
- Listener-specified
 - ExpFD: Fréchet distance in expression coefficients from 3DMM reconstruction
 - PoseFD: Fréchet distance in translation/rotation coefficients from 3DMM reconstruction

The official Github code repository of the competition provides the code for calculating these metrics³, enabling us to effectively evaluate these metrics locally as well.

²<https://vico.solutions/challenge/2023>

³https://github.com/dc3ea9f/vico_challenge_baseline/tree/main/evaluations

2.3 Constructing the Validation Set

Constructing a validation set is a crucial step in the process. It allows for the computation of various metrics locally, enabling targeted improvements to be made. This approach facilitates quick iterations and evaluation of enhancements made to metrics that previously underperformed, without the need to wait for one to two hours after submission on the competition website.

The validation set we had constructed was a subset selected from the training set, specifically consisting of a total of 28 UUIDs from different video IDs belonging to speaker IDs 37, 30, 51, 58, and 71. The rationale behind selecting these particular speaker IDs for the validation set was twofold. Firstly, they were the most frequently occurring speaker IDs in the training set, making them more representative and generalizable. Secondly, speaker ID 37 also appeared in the test set, thus including it in the validation set ensured better representation of the test set.

This method of constructing a validation set ensured that we could quickly compute various metrics locally and that the computed results closely approximated the online metrics scores of the test set.

2.4 Improvements for Vivid Talking Head Video Generation

2.4.1 Direct Inference with SadTalker. In our first attempt, we input the test dataset of this track, which included the first frames and corresponding speech audios of 45 speakers, into the inference code "inference.py" of SadTalker. Except for setting the "still" argument, all other arguments were kept as default. The generated videos were then submitted as the results to the competition website.

Since SadTalker came with pre-trained models for each stage, we were able to directly perform inference without the need to train the models using the training dataset. In fact, throughout the entire competition process, we did not train or fine-tune any models.

The video generation results of SadTalker, with minimal modifications to the default arguments, performed well on the leaderboard and reached the third position two days before the end of the competition. This success provided ample support and confidence for further improvements.

The online evaluation metrics for the initial submission indicated that the SadTalker framework, with almost default arguments, performed well in terms of naturalness and speaker-specified metrics. However, it showed poor performance in terms of visual quality. Therefore, our subsequent focus was primarily on improving the visual quality aspect.

2.4.2 Removing Preprocessing. SadTalker has several preprocessing modes, corresponding to the "preprocess" argument in "inference.py". The three main preprocessing modes are crop, resize, and full:

- **crop:** This mode crops only the face portion from the image, regardless of its size. Subsequent operations such as 3DMM and video generation are performed on the cropped face.
- **resize:** After cropping the face, it is resized to a specified size before performing operations such as 3DMM and video generation.

- **full**: After cropping the face, a video is generated for the cropped portion and then pasted back into the original image at the location of the face. This approach may appear unnatural if the image is not still.

These preprocessing modes provide different ways to manipulate the input image and perform subsequent operations on the face region.

However, the dataset for this challenge is actually quite well-organized. On one hand, the size of the images in the dataset is 256 by 256, which coincidentally matches the required size for SadTalker’s pre-trained models, so cropping is unnecessary. On the other hand, the faces in the dataset are also positioned relatively consistently, mostly near the center of the images, and there is not a significant variation in face sizes. These factors indicate that there is no need for preprocessing, and the 3DMM can be directly generated from the images. Some samples from the dataset are shown in Figure 1 as illustrated.



Figure 1: Several sample images are provided to demonstrate that the faces in the dataset are relatively standardized.

After removing the preprocessing stage of SadTalker and directly proceeding to the subsequent steps, there was indeed a significant improvement in visual quality-related metrics, while other metrics did not show any noticeable decline.

2.4.3 Trying Enhancers. In general, it usually applies some post-processing techniques such as gfpgan[2] to enhance the generated videos by SadTalker. In fact, this kind of enhancement has become almost standard in any facial generation-related applications because although it significantly increases processing time, it does improve the quality of the generated results.

Initially, it was believed that applying enhancement would increase the metrics. However, actual testing showed the opposite,

as it actually decreased the scores. This is because visual quality metrics such as PSNR and SSIM require the generated image to be as close as possible to its ground truth. However, these types of post-processing enhancements for facial generation result in larger differences between the generated images and the ground truths, without providing any benefits in other metrics. Therefore, when using SadTalker to generate videos for this challenge, the enhancer argument should not be set.

2.4.4 Adding Matting as Post-processing. After the aforementioned improvements, the visual quality metrics fell short of expectations, and there was a significant gap compared to the leading teams. Upon observation, we noticed a discrepancy between the generated frames’ background (outside the face) and the actual background. Although not visible to the naked eye, there was a slight deviation in each pixel of the background, which had a considerable impact on visual quality.



Figure 2: Several sample images are provided to demonstrate the computed alpha mattes by MODNet and the results of image matting.

To address this issue, we employed MODNet⁴ [1] for matting the faces and used the alpha values obtained from pretrained MODNet to weight the first frame and the generated frames separately. Some examples of image matting computed by MODNet are shown in Figure 2. It can be observed that although the results are not particularly flawless, they are already sufficient for the purpose at hand.

By summing the weighted frames, we merged the background of the first frame with the foreground of the generated frames, as shown in Equation 1:

$$F_f = \alpha F_g + (1 - \alpha) F_0 \quad (1)$$

⁴<https://github.com/ZHKKKe/MODNet>

Table 1: Talking Head Generation

	SSIM↑	CPBD↑	PSNR↑	FID↓	CSIM↑	PoseL1↓	ExpL1↓	AVOffset→	AVConf↑	LipLMD↓
direct inference	0.583	0.161	17.108	31.306	0.508	0.100	0.150	-0.867	1.978	11.849
wo cropping, wo matting	0.590	0.168	17.390	33.316	0.521	0.094	0.147	-1.200	1.948	11.402
wo cropping, w matting	0.613	0.203	17.822	29.163	0.546	0.101	0.147	-0.867	2.564	12.147

Table 2: Listening Head Generation

	SSIM↑	CPBD↑	PSNR↑	FID↓	CSIM↑	PoseL1↓	ExpL1↓	PoseFD↓	ExpFD↓
static	0.611	0.209	19.291	36.809	0.682	0.075	0.150	0.017	0.741
wo cropping, wo matting	0.610	0.180	18.974	30.943	0.676	0.079	0.149	0.027	0.603
wo cropping, w matting	0.612	0.186	19.064	30.982	0.676	0.078	0.149	0.024	0.604

where F_f is the frame that serves as the final generated result, F_g is the frame generated by SadTalker, and F_0 is the first frame provided by the dataset.

It should be noted that post-processing with matting can improve visual quality, but the videos in the dataset must meet the following prerequisites:

- stationary camera shots
- unchanged background
- minimal movement of the subjects' heads

Otherwise, the fusion result may exhibit chaotic backgrounds and foregrounds.

Fortunately, the dataset for this challenge satisfies these conditions. The evaluation on the validation set indicated that this approach significantly enhanced various metrics related to visual quality, while other metrics did not show significant decline. The online scores after submission also demonstrated noticeable improvement.

2.5 Improvements for Responsive Listening Head Video Generation

We made relatively few attempts on this track, as it proved to be more difficult than the Talking Head track. Despite submitting only three times, we were fortunate enough to achieve second place. However, it is important to note that the evaluation metrics were somewhat lacking, and our success cannot be solely attributed to luck.

After several submissions for the Talking Head track, we began working on this track with a better understanding of which methods might yield better results. The techniques we used here were similar to those employed in the Talking Head track. Let us briefly introduce the methods we utilized for this track.

Firstly, we attempted a simple baseline method where all frames were replaced with the first frame, resulting in a static video. Surprisingly, the scores obtained from submitting these results were good, highlighting the flaws in the evaluation metrics.

Next, we employed the same method as the Talking Head track, using SadTalker for listening head generation. Although this approach might not have been entirely reasonable, as the generated

videos depicted the listener's mouth moving, it resulted in an improvement in scores. However, it was important to acknowledge that the generated videos still fell short of capturing the expected reactions of the listener.

3 EXPERIMENTS

Table 1 presents the scores obtained from different approaches discussed in Sections 2.4.1, 2.4.2, and 2.4.4. The first row corresponds to the scores inferred directly using the default arguments of SadTalker, as mentioned in Section 2.4.1. The second row represents the scores obtained after removing preprocessing steps, as discussed in Section 2.4.2. Lastly, the third row displays the scores achieved by incorporating matting as a post-processing technique, as described in Section 2.4.4.

It can be observed that, overall, the first five metrics representing visual quality demonstrate a consistent improvement, with the version featuring matting post-processing showing the most significant enhancement. Meanwhile, the other metrics do not exhibit a noticeable decrease.

In Table 2, the first row represents a static video generated using only the first frame. The second and third rows correspond to the methods used in the corresponding rows of Table 1.

The improvements in the results of the listening head are not as significant as those in the talking head, because strictly speaking, the methods used are not as reasonable for listening head generation.

4 FUTURE WORK

We believe that the immediate priority is to design better evaluation methods in order to minimize the gap between human judgment and machine judgment, particularly for listeners. It would be beneficial if metrics could be developed that align speech content with facial expressions, gestures, and other non-verbal cues. In this regard, [6] has made some valuable explorations, and we hope to apply them in the next competition.

Additionally, we feel that the videos in the dataset are somewhat simplistic. If the backgrounds could be more complex and if evaluations could encompass not only the head but also other body parts, and even body movements corresponding to the speech content, it would greatly enhance the value of the generation task.

REFERENCES

- [1] Zhanghan Ke, Jiayu Sun, Kaican Li, Qiong Yan, and Rynson WH Lau. 2022. Modnet: Real-time trimap-free portrait matting via objective decomposition. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 36. 1140–1147.
- [2] Xintao Wang, Yu Li, Honglun Zhang, and Ying Shan. 2021. Towards real-world blind face restoration with generative facial prior. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 9168–9178.
- [3] Wenxuan Zhang, Xiaodong Cun, Xuan Wang, Yong Zhang, Xi Shen, Yu Guo, Ying Shan, and Fei Wang. 2023. SadTalker: Learning Realistic 3D Motion Coefficients for Stylized Audio-Driven Single Image Talking Face Animation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 8652–8661.
- [4] Mohan Zhou, Yalong Bai, Wei Zhang, Ting Yao, and Tiejun Zhao. 2023. Interactive Conversational Head Generation. *arXiv preprint arXiv:2307.02090* (2023).
- [5] Mohan Zhou, Yalong Bai, Wei Zhang, Ting Yao, Tiejun Zhao, and Tao Mei. 2022. Responsive listening head generation: a benchmark dataset and baseline. In *European Conference on Computer Vision*. Springer, 124–142.
- [6] Mohan Zhou, Yalong Bai, Wei Zhang, Ting Yao, Tiejun Zhao, and Tao Mei. 2023. Learning and Evaluating Human Preferences for Conversational Head Generation. *arXiv preprint arXiv:2307.10636* (2023).